Cloud Computing – Beginner Notes

1. Introduction to Cloud Computing

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, and software over the internet. Instead of buying and maintaining all the hardware locally, users can access computing resources when they need them and often pay according to usage.

Examples of cloud-based services include:

- Online file storage and backup
- Web hosting and application hosting
- Online databases
- Video and collaboration platforms
- Data processing and analytics

2. Characteristics of Cloud Computing

On-demand self-service: Users can obtain computing resources when required without manually setting up physical hardware.

Broad network access: Services can be accessed through networks using devices such as computers, tablets, and smartphones.

Resource pooling: A cloud provider can share physical resources among multiple customers while logically separating their environments.

Rapid elasticity: Resources can often be increased or decreased according to demand.

Measured service: Cloud usage can be monitored and measured, supporting usage-based billing and resource management.

3. Cloud Service Models

Infrastructure as a Service (IaaS): Provides fundamental computing resources such as virtual machines, storage, and networks. The customer manages much of the operating system and application environment.

Platform as a Service (PaaS): Provides a managed platform for developing and running applications. Developers can focus more on application code while the provider manages much of the underlying infrastructure.

Software as a Service (SaaS): Provides ready-to-use software through the internet. Users generally access the application without managing the underlying servers or operating system.

4. Cloud Deployment Models

Public Cloud: Cloud infrastructure is provided by a third-party provider and made available to customers over a network. It can be useful for organizations that want scalable resources without owning all the physical infrastructure.

Private Cloud: Cloud infrastructure is dedicated to a single organization. It can provide greater control over the environment, depending on how it is designed and managed.

Hybrid Cloud: Combines private and public cloud environments, allowing data or applications to use different environments when appropriate.

Community Cloud: Infrastructure is shared by organizations with common requirements or goals. Its use depends on the needs and governance arrangements of the participating organizations.

5. Advantages of Cloud Computing

- **Cost efficiency:** Reduces the need for large upfront hardware investments.
- **Scalability:** Resources can be expanded or reduced as requirements change.
- **Accessibility:** Services can often be accessed from different locations and devices.
- **Backup and recovery:** Cloud services can support data backup and disaster recovery strategies.
- **Faster deployment:** Computing resources can often be provisioned faster than purchasing physical equipment.
- **Collaboration:** Cloud-based applications can allow teams to work with shared resources.

6. Disadvantages and Challenges

- Dependence on reliable network connectivity
- Security and privacy responsibilities must be carefully managed
- Possible service outages or provider dependency
- Costs can increase if resources are not monitored and controlled
- Moving applications between providers may require planning and technical work

7. Common Cloud Technologies

Virtualization allows multiple virtual environments to run on physical hardware. Containers package an application and its dependencies into a portable unit. Cloud databases provide managed storage and database services. Serverless computing allows developers to run functions without directly managing servers.

8. Cloud Computing and Virtualization

Virtualization is an important technology used in many cloud environments. A physical server can be divided into multiple virtual machines, with each virtual machine running its own operating system and applications. This helps providers use hardware efficiently and allocate resources flexibly.

9. Cloud Security

Cloud security involves protecting data, applications, accounts, and infrastructure. Important practices include strong authentication, access control, encryption, regular updates, monitoring, backups, and careful management of permissions. Security is a shared responsibility: the cloud provider protects parts of the underlying service, while the customer is responsible for configuring and using services securely.

10. Examples of Cloud Platforms

Major cloud platforms include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud. These platforms provide services for computing, storage, databases, networking, analytics, artificial intelligence, application development, and many other workloads.

11. Cloud Computing in Daily Life

People use cloud computing every day, often without thinking about it. Examples include storing photos online, using web-based email, streaming media, collaborating on online documents, and backing up mobile data. Businesses use cloud platforms to host websites, run applications, analyze data, and support remote work.

12. Basic Cloud Computing Workflow

1. Choose the required cloud service.
2. Create and configure the required resources.
3. Deploy applications or store data.
4. Set access controls and security settings.
5. Monitor performance, usage, and costs.
6. Scale resources when demand changes.
7. Back up important data and review the environment regularly.

13. Conclusion

Cloud computing provides flexible access to computing resources over the internet. Its main service models are IaaS, PaaS, and SaaS, while common deployment approaches include public, private, and hybrid cloud. Learning cloud fundamentals, virtualization, security, networking, and cost management provides a strong foundation for students and beginners who want to enter cloud technology or IT careers.